

01 Equipment overview

Fictional equipment manual - demo use only.
Manufacturer, model, fault codes and technical behaviour in this document are invented for a software demonstration. This document is not approved for operating, diagnosing or repairing real equipment.

Document control	Value
Manufacturer	Northline Example Systems (fictional)
Equipment model	MCV-200 modular conveyor
Document number / type	NX-MCV200-OM-001 / Operation Manual
Version / issue date	1.0 / 12 September 2026
Applicability	MCV-200 with fictional controller MC-20, configuration DEMO-A

1.1 Intended use and assembly

The MCV-200 represents a powered conveyor for moving closed, dry cartons between packaging stations. The fictional assembly includes a conveyor belt, drive motor, guarded drive transmission, entry sensor, operator display and safety interlock circuit. The operator display identifies the active fault code.

Equipment names such as "Conveyor 3" identify individual installations. They are not model names. Confirm the model and controller configuration before applying a fault entry; this manual does not cover other conveyor models or controller variants.

1.2 Document map

Page	Subject
2	Operating context, safety prerequisites and observation boundaries
3	Fault-code reference and fault identification
4	E-402: motor thermal-protection input active
5	E-215 and E-310: other fault guidance
6	Maintenance reporting, escalation and document control

02 Operating context & prerequisites

2.1 Applicability before troubleshooting

Confirm the equipment ID, model MCV-200 and controller configuration DEMO-A. Read the full displayed fault code and record whether the machine is stopped or operating intermittently. A verbal description such as "motor broken" does not identify a confirmed component failure.

Use the site-approved procedures for any real installation. The example conveyor is described as handling closed, dry cartons within its approved installation envelope. No numeric load, temperature, speed or reset limits are specified in this demo manual; do not infer them from the model name.

Prerequisite for every fault entry: Follow the applicable site emergency and hazardous-energy-control procedures. Physical inspection, guard removal, electrical access, blockage removal and repair require appropriately authorized personnel and the prescribed safe condition. A stopped conveyor is not evidence of energy isolation.

2.2 Observation boundaries

An operator may report information already available from the normal operator position: the display code, time of occurrence, whether movement stopped and conditions noticed before the stop. Do not reach into the conveyor, touch a motor to assess temperature, remove guards or open an electrical enclosure to gather information for a report.

If smoke, a burning smell, exposed electrical parts or another immediate hazard is observed, follow the site emergency procedure and notify the responsible personnel. Do not delay the response to complete documentation.

2.3 Actions requiring maintenance ownership

Checks of the thermal-protection circuit, drive mechanism, sensor alignment or safety interlocks belong to authorized maintenance personnel. The applicable site procedure determines safe access, diagnostic methods and return-to-service authorization. Never bypass an interlock or repeatedly reset a fault to resume production.

2.4 Information to retain

Preserve the displayed code and the original observations before the information is lost. Record actions only if someone confirms they were performed. Recommended checks in this manual must not be recorded as completed work. If the cause is not established, state "Cause not yet confirmed."

03 Fault-code reference

The following codes apply only to the fictional MCV-200 / MC-20 DEMO-A configuration. Each code identifies a reported condition, not a verified root cause. All entries require the prerequisites on page 2.

Code	Displayed condition	Meaning and reference
E-402	Motor thermal protection active	The controller reports an active motor thermal-protection input. It does not establish whether the motor is damaged. See page 4.
E-215	Entry sensor signal inconsistent	The controller cannot reconcile the entry sensor signal with the expected carton-entry sequence. See page 5, section 5.1.
E-310	Guard interlock open	The controller reports an open guard-interlock circuit and inhibits movement. See page 5, section 5.2.

3.1 Record the displayed condition precisely

Copy the fault code from the operator display. Record any accompanying text and whether other codes appeared. When several faults are reported, preserve their sequence if known. Do not choose a fault code solely because its description sounds similar to a symptom.

3.2 Identify the right reference

Use the exact fault entry for the confirmed model and configuration. For E-402, read page 4 together with the prerequisites on page 2. For E-215 or E-310, use the appropriate subsection on page 5 together with page 2. Reporting requirements are on page 6.

3.3 Unlisted faults or missing code

If the code is absent from this table, the manual provides no code-specific explanation. Record the code exactly, capture the observed symptoms and refer the issue to maintenance. If no code is displayed, report that fact instead of assigning one. Similar wording in another fault entry is not sufficient to establish the meaning of an unlisted code.

Interpretation boundary: A code match supports retrieval of the relevant guidance. It does not prove that a listed possible cause is present or that a listed corrective action is appropriate.

04 E-402: motor thermal protection

Fault E-402 | Display: Motor thermal protection active
Applies to MCV-200 / MC-20 DEMO-A. Prerequisites: page 2. Reporting: page 6.

4.1 Documented meaning

E-402 indicates that the controller is receiving an active motor thermal-protection input. In this fictional configuration, the conveyor drive is inhibited while that input is active. The indication can be consistent with motor heating, but the code alone does not confirm the actual motor temperature, permanent motor damage or the underlying cause.

4.2 Possible causes - not confirmed findings

Possible cause	Information relevant to investigation
Restricted motor cooling	Obstruction near the cooling path or a change in local ventilation may affect cooling.
Elevated mechanical load	A change in carton loading or resistance in the drive system may increase motor demand.
Thermal-protection signal fault	A protection device or its signal path may require investigation even if heating has not been established.

4.3 Operator observations to capture

Record the time E-402 first appeared, whether the conveyor stopped, and whether the issue recurred during the shift. Describe any change in loading or operating conditions noticed before the event. Record unusual noise or smell only if actually observed. Do not touch the motor, access guarded areas or restart the conveyor to reproduce the event.

4.4 Recommended maintenance investigation

After the page 2 prerequisites are satisfied, authorized maintenance personnel should assess the cooling path, mechanical loading and thermal-protection signal using the applicable approved procedures. These are investigation areas, not a confirmed diagnosis or a prescribed repair sequence. Do not select a replacement motor solely from E-402.

4.5 Escalation and report wording

An unresolved or recurring E-402 requires maintenance review before any authorized return to service. Immediate hazards follow the site emergency procedure. A report may state: "E-402 displayed; conveyor reported stopped; cause not yet confirmed." Include operating conditions and completed checks only when supplied by the reporting person.

05 Other fault guidance

Applies to MCV-200 / MC-20 DEMO-A. Read the prerequisites on page 2 before using either entry. A reported sensor or interlock condition must not be treated as permission to bypass a protective function.

5.1 E-215: entry sensor signal inconsistent

Documented meaning. The controller cannot reconcile the entry sensor signal with the expected carton-entry sequence. The event does not itself establish that the sensor has failed.

Possible causes. A carton or other object may be obstructing the sensing area; the sensor may be misaligned or contaminated; or the signal path may require investigation. These alternatives need confirmation by maintenance.

Operator observations. Record when E-215 appeared, whether cartons accumulated at the entry and whether there was a recent change in carton presentation. Describe only what can be observed from the normal operator position. Do not reach into the entry area or reposition a sensor to gather evidence.

Recommended check and escalation. Authorized personnel should investigate the sensing area, alignment and signal under the approved site procedure. An unresolved or recurring condition requires maintenance review. Record the symptoms; do not state "sensor replaced" or "sensor defective" without evidence.

5.2 E-310: guard interlock open

Documented meaning. The controller reports an open guard-interlock circuit and inhibits movement. This may represent an open guard or an issue requiring investigation; it does not prove that a switch is defective.

Possible causes. A guard may not be in its intended closed position, or an interlock component or its signal path may require assessment. The code does not identify which of these conditions is present.

Operator observations. Record the code, time and any guard state visible without approaching a hazard. Note whether maintenance activity was already in progress if known. Do not manipulate guards or operate the machine merely to test the report.

Recommended check and escalation. The responsible authorized personnel should investigate using the site-approved safety procedure. Never bridge, bypass or defeat the interlock. Return to service requires the applicable authorization; acknowledging a report does not provide that authorization.

06 Reporting & escalation

6.1 Minimum breakdown-report information

Field	What to record
Equipment	Internal equipment ID, name, model and location. Do not substitute the model for the equipment ID.
Observed condition	Displayed fault code and the reporting person's symptoms; distinguish a stopped machine from intermittent operation.
Timing and context	When the issue was first noticed, recurrence if known and relevant changes in operating conditions.
Completed checks	Only actions confirmed by the reporting person. Leave unknown actions unconfirmed.
References	Manual NX-MCV200-OM-001, version 1.0; relevant fault page and prerequisites on page 2.
Outstanding information	Unknown cause, unanswered questions and the responsible maintenance contact or team, if known.

6.2 Separate evidence from guidance

Preserve the original report text. A prepared description may organize and clarify those observations, but it must not add an unobserved temperature, failed component, repair, elapsed downtime or completed inspection. Put possible causes and recommended checks in a separate reference-guidance area. State uncertainty explicitly when a finding is not established.

6.3 Handover and return to service

Route the report using the site's maintenance process. The responsible planner or maintenance team determines assessment, priority and subsequent work planning. Saving or acknowledging a breakdown report does not authorize repair, restart or release of equipment. Immediate hazards follow the site emergency procedure rather than waiting for report completion.

6.4 Routine inspection records

Inspection frequency and methods are governed by the approved site maintenance plan; no intervals are prescribed here. Record the date, equipment ID, authorized person, actual observations and follow-up reference. Prior inspection records can provide context but do not establish the cause of a new event.

6.5 Version and scope limits

Revision 1.0 is the initial fictional issue dated 12 September 2026. Keep this PDF unchanged when retaining references; issue a new version for revised content. This six-page document contains no spare-part catalogue, electrical schematics, repair instructions or numeric reset criteria. Do not infer missing specifications.